Lecture 11: Quadratic approximation and linear predictions

- So far, we have seen the grid approximation. It transforms a continuous parameter space into a discrete one.
- More annoyingly, it becomes computationally prohibitive when the number of parameters reaches three or four. Think of simple linear regression with an intercept, a slope and a standard deviation for the normal residuals. And now of a multivariate regression with 10 predictors!?!

- A quadratic (or Laplace) approximation will approximate the posterior as a Gaussian distribution.
- We can estimate the MAP (Maximum A Posteriori), which will also be the mean/median, and the standard deviations/correlations among parameters.
- With flat priors, we the Bayesian analysis is very close to maximum likelihood estimation. Same numbers, different meanings.

Multivariate Normal Distribution

Introducing Random Vectors

- The multivariate normal distribution is very convenient for modelling dependent continuous rvs.
- To describe it, we collect together all rvs in a **random vector**:

$$\boldsymbol{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

  - Conventional to organize rvs as a column vector.
- Then

$$\mathbb{E}[\boldsymbol{X}] = \mathbb{E}\begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} = \boldsymbol{\mu} \qquad \text{where } \mu_i = \mathbb{E}[X_i].$$

Multivariate Normal Distribution

Covariance Matrices

- The variances and covariances for $X_1, \ldots, X_n$ are organized in the **covariance** (or **dispersion**) matrix for $\boldsymbol{X}$:

$$\text{Var}(\boldsymbol{X}) = \text{Var}\begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_n^2 \end{bmatrix}$$

- Covariance matrix has variances on long-diagonal, covariances elsewhere.
- Notice that: $\sigma_{ij} = \text{Cov}(X_i, X_j) = \text{Cov}(X_j, X_i) = \sigma_{ji}$, so $\text{Var}(\boldsymbol{X})$ is a **symmetric** matrix.
- Reminder:

$$\text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j]$$

where $\mu_i = \mathbb{E}[X_i]$ and $\mu_j = \mathbb{E}[X_j]$.

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Multivariate Normal Distribution

Density Function

- A random vector $\boldsymbol{X}$ has a multivariate normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix Σ if its joint probability density function is given by

$$f(\boldsymbol{x}) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} e^{-(\boldsymbol{x}-\boldsymbol{\mu})^{\mathsf{T}}\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu})/2}$$

  - In this expression, $|\Sigma|$ indicates the determinant of the matrix Σ.
- Denoted by $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{\mu}, \Sigma)$.
- Probabilities can be computed as volumes under the density function.
- The marginal distribution of X_i is then $\mathrm{N}(\mu_i, \sigma_i^2)$.

Example: A Bivariate Normal Distribution

Let $\boldsymbol{X} = (X_1, X_2)^\mathsf{T} \sim \mathrm{N}(\boldsymbol{\mu}, \Sigma)$ with

$$\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{X}] = \begin{bmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

and

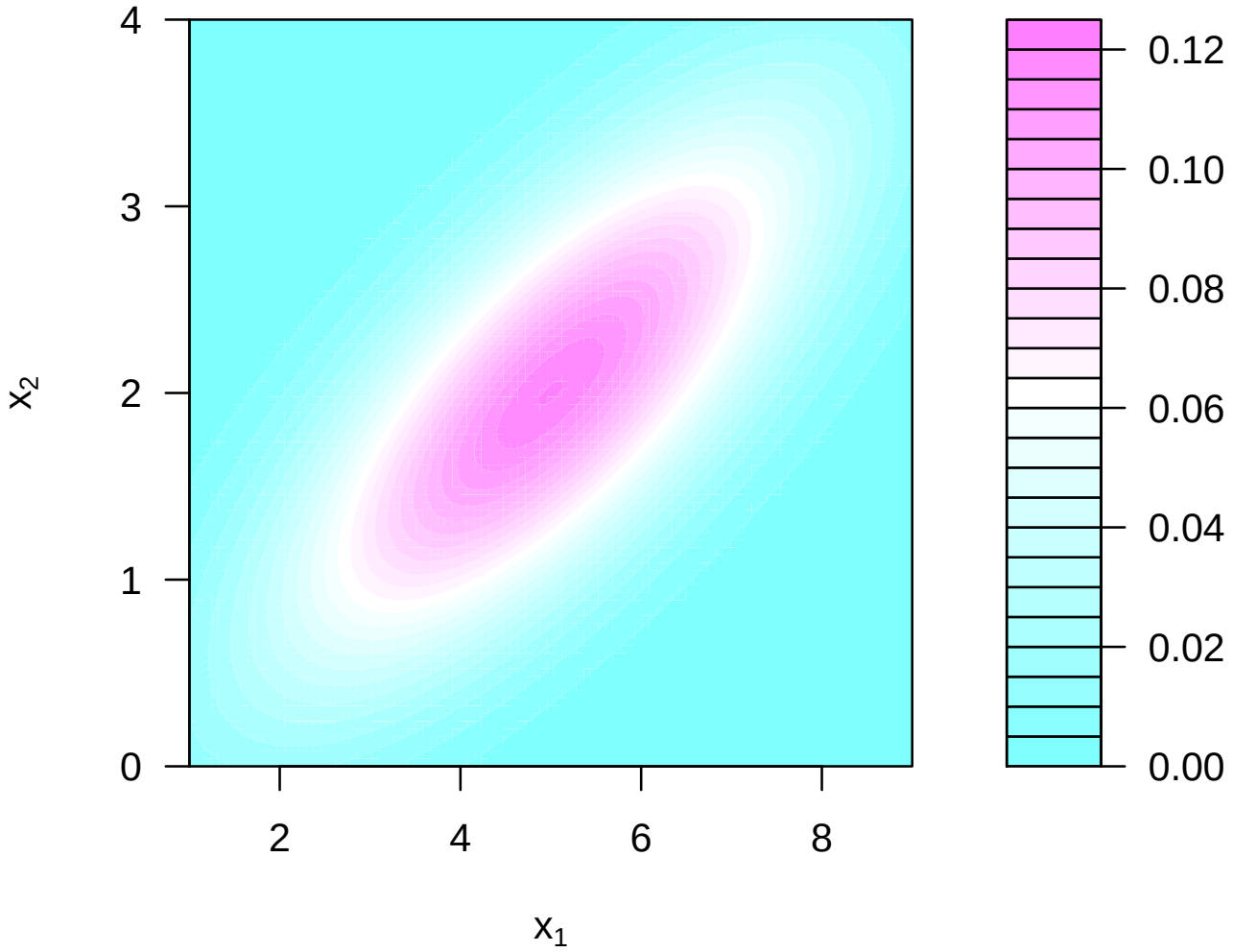
$$\Sigma = \mathrm{Var}(\boldsymbol{X}) = \mathrm{Var} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{bmatrix} = \begin{bmatrix} 4 & 1.5 \\ 1.5 & 1 \end{bmatrix}$$

Then

$$\mathrm{Cor}(X_1, X_2) = \rho = \frac{\sigma_{12}}{\sigma_1 \sigma_2} = \frac{1.5}{2 \times 1} = 0.75$$

Example continued

Plot of probability density function



Using `quap()`

```
flist <- alist(
height ~ dnorm(mu, sigma)
mu ~ dnorm(178, 20)
sigma ~ dunif(0,50)
)
```

$$h_i \sim \text{Normal}(\mu.\sigma)$$
$$\mu \sim \text{Normal}(178, 20)$$
$$\sigma \sim \text{Unif}(0, 50)$$

And then simply:

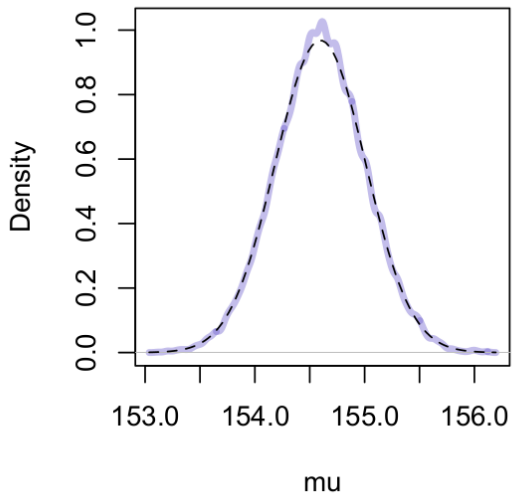
```
m4.1 <- quap(flist, data = d2)
precis(m4.1) %or use summary
```

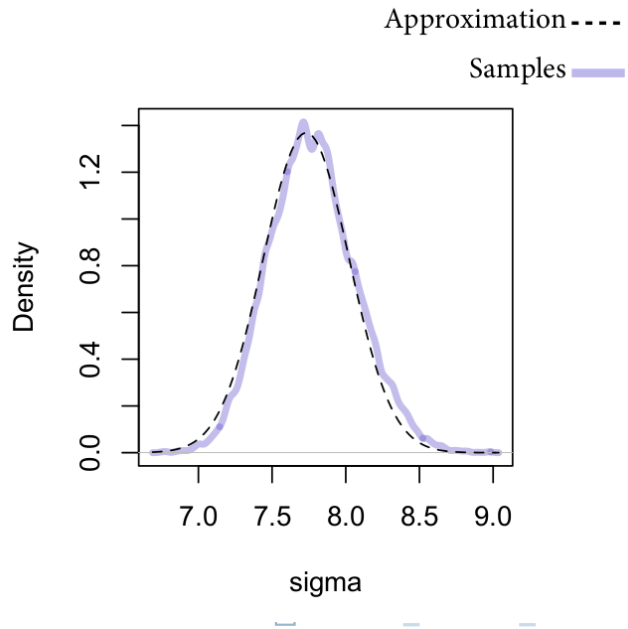
```
       Mean StdDev   5.5%  94.5%
mu   154.61   0.41 153.95 155.27
sigma  7.73   0.29   7.27   8.20
```

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Extracting samples from a the `quap` approximation model

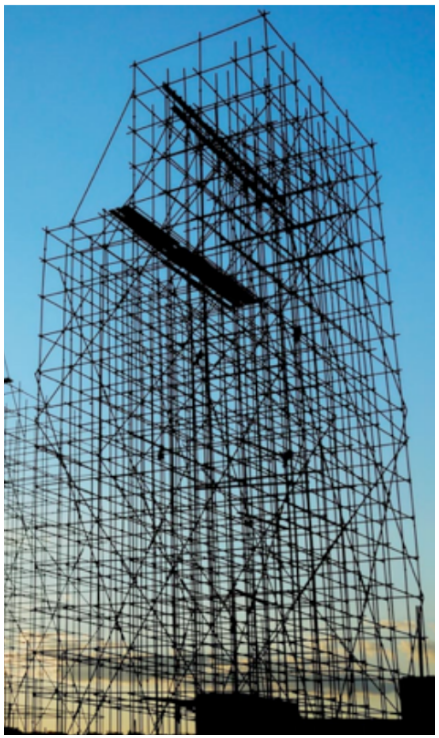
```
post <- extrac.samples(m4.1, n = 10000)
head(post, 4)
        mu      sigma
1 155.0031 7.443893
2 154.0347 7.771255
3 154.9157 7.822178
4 154.4252 7.530331
```





`quap` - scaffold

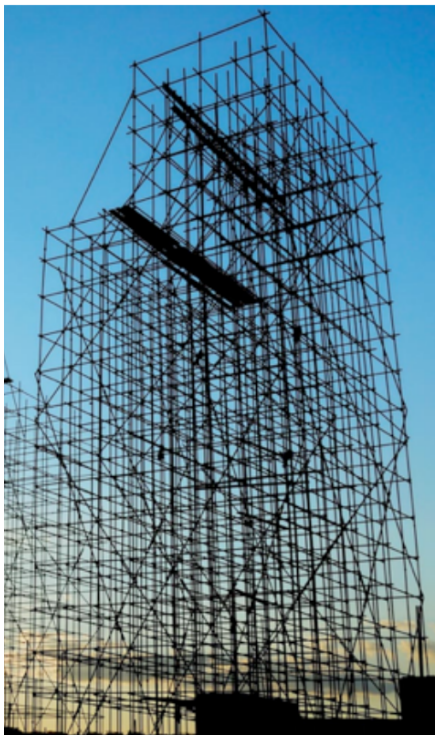
- `quap` is a scaffold
- It forces a full specification of the model, so it is very instructive and forces you to think.
- It is very efficient, quick and works with a wide class of models.
- Same as *penalised maximum likelihood*
- Not always a good way to approximate the prior, in particular when the normal approximation is not ideal



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`quap` - scaffold

- `quap` is a scaffold
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- It is very efficient, quick and works with a wide class of models.
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- Not always a good way to approximate the prior, in particular when the normal approximation is not ideal

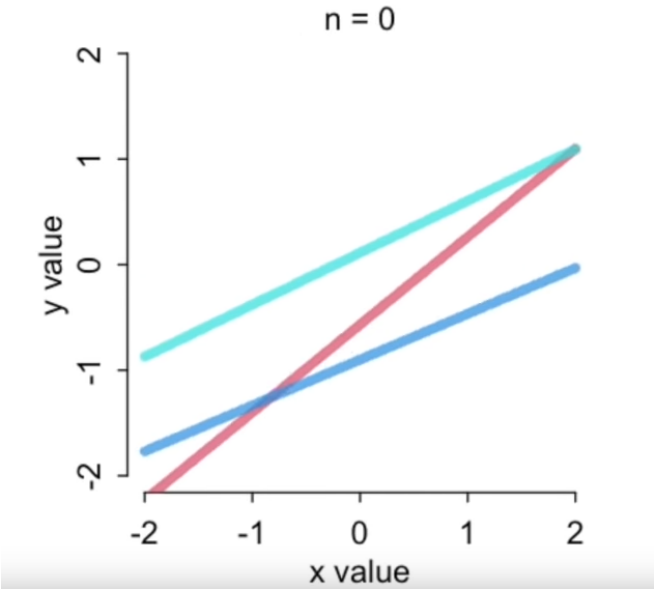


- Maybe rethink about priors, etc.

Statistical model

Thinking about priors

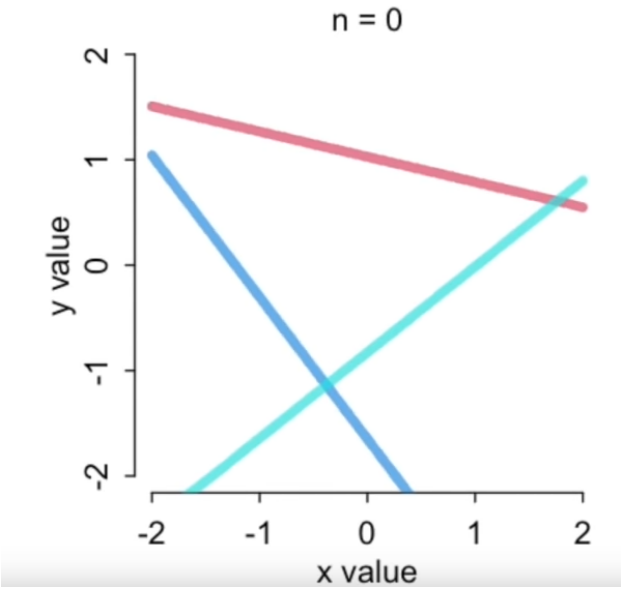
$$
\begin{aligned}
y_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta x_i \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Unif}(0, 1)
\end{aligned}
$$



Statistical model

Thinking about priors

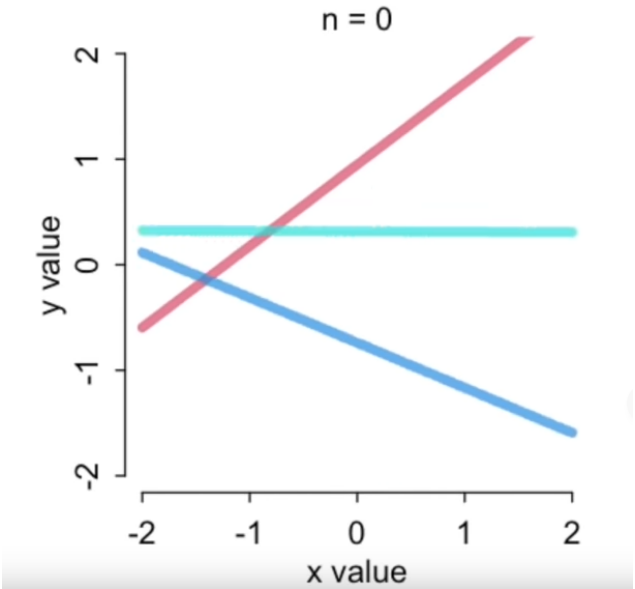
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\begin{aligned}
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\mu_i &= \alpha + \beta x_i \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Unif}(0, 1)
\end{aligned}
$$



Statistical model

Thinking about priors

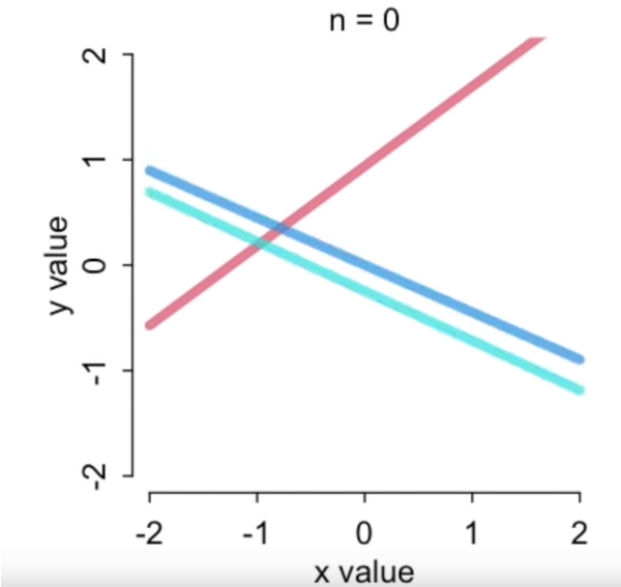
$$
\begin{aligned}
y_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta x_i \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Unif}(0, 1)
\end{aligned}
$$



Statistical model

Thinking about priors

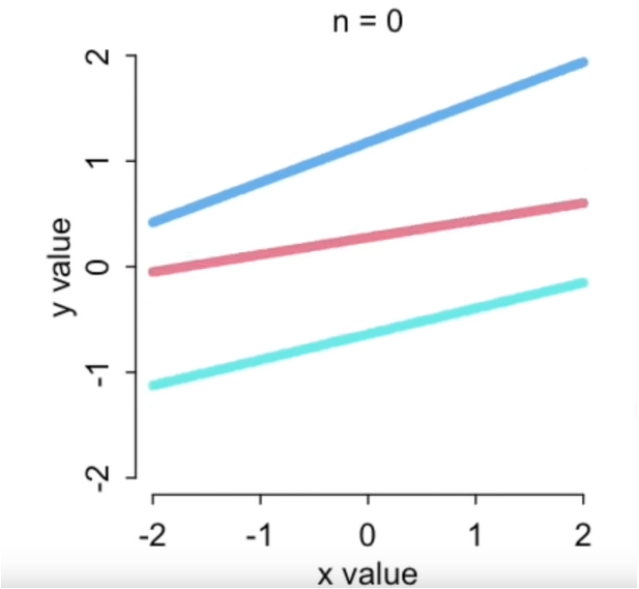
$$
\begin{aligned}
y_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta x_i \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Unif}(0, 1)
\end{aligned}
$$



Statistical model

Thinking about priors

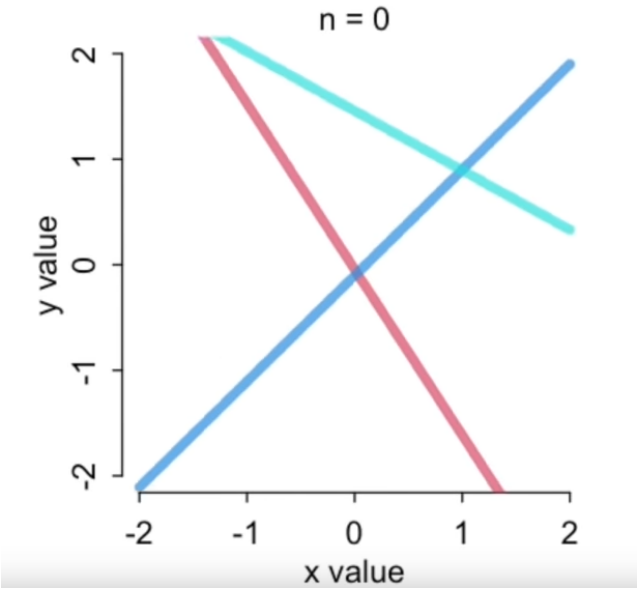
$$
\begin{aligned}
y_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta x_i \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Unif}(0, 1)
\end{aligned}
$$



Statistical model

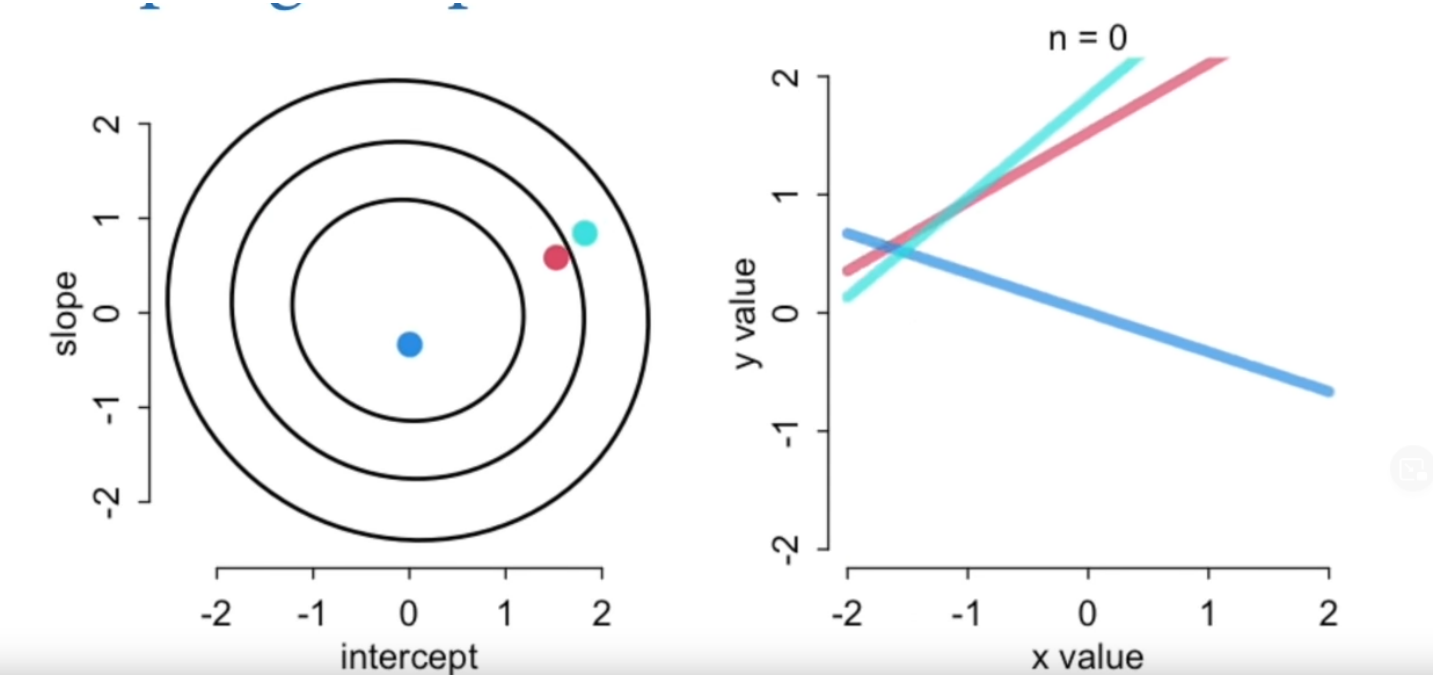
Thinking about priors

$$
\begin{aligned}
y_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta x_i \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Unif}(0, 1)
\end{aligned}
$$



Statistical model

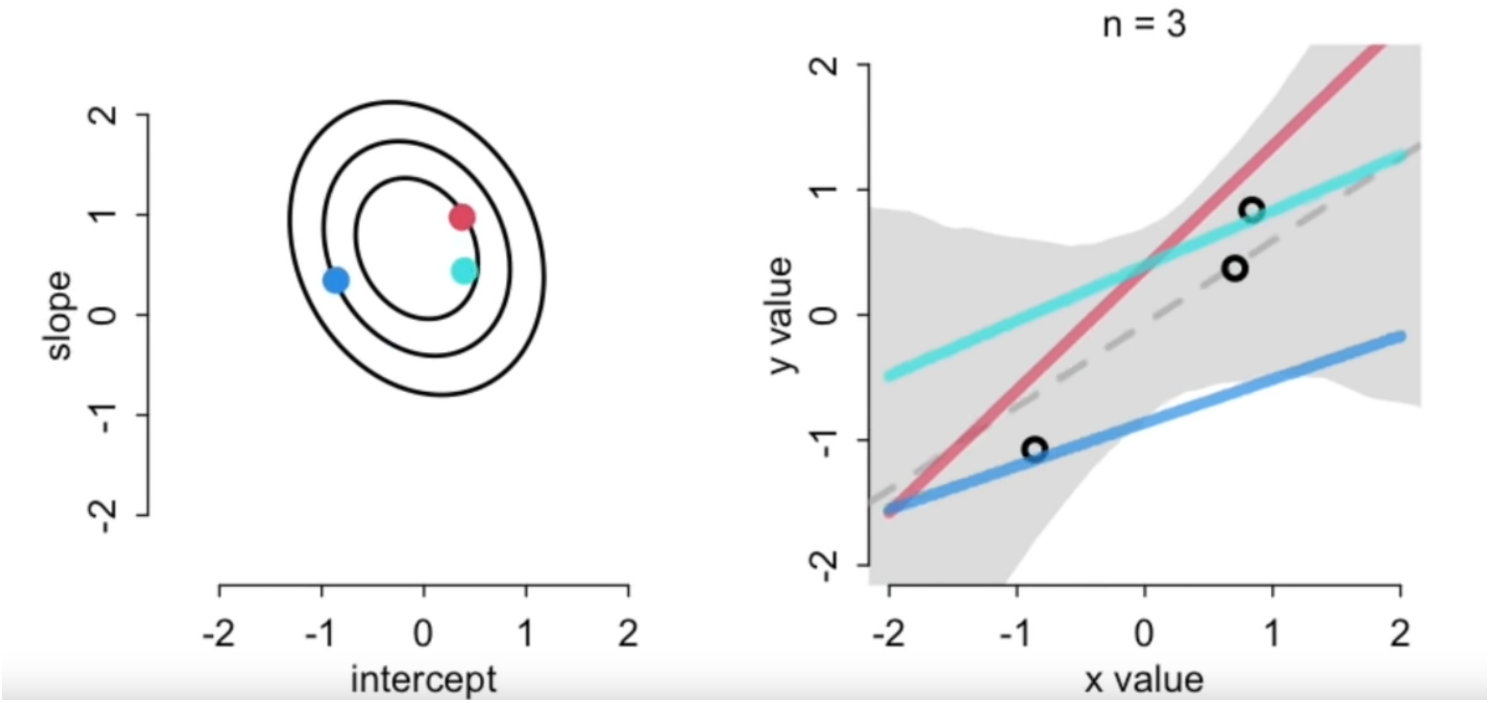
Thinking about priors



Do such simulations! (using `rnorm()` and `runif()`)

Statistical model

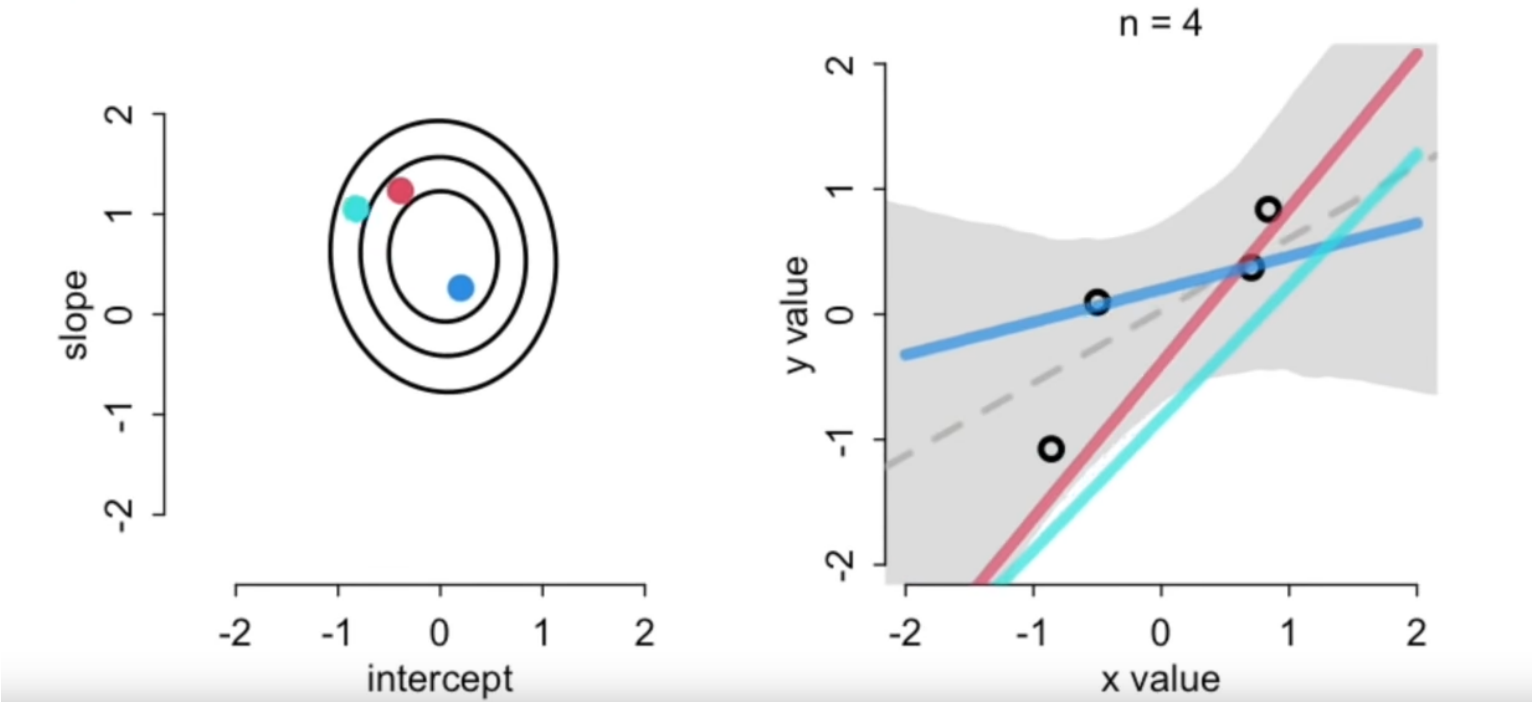
Updating the posterior



Grey regions $\approx$ highest posterior density line regions.

Statistical model

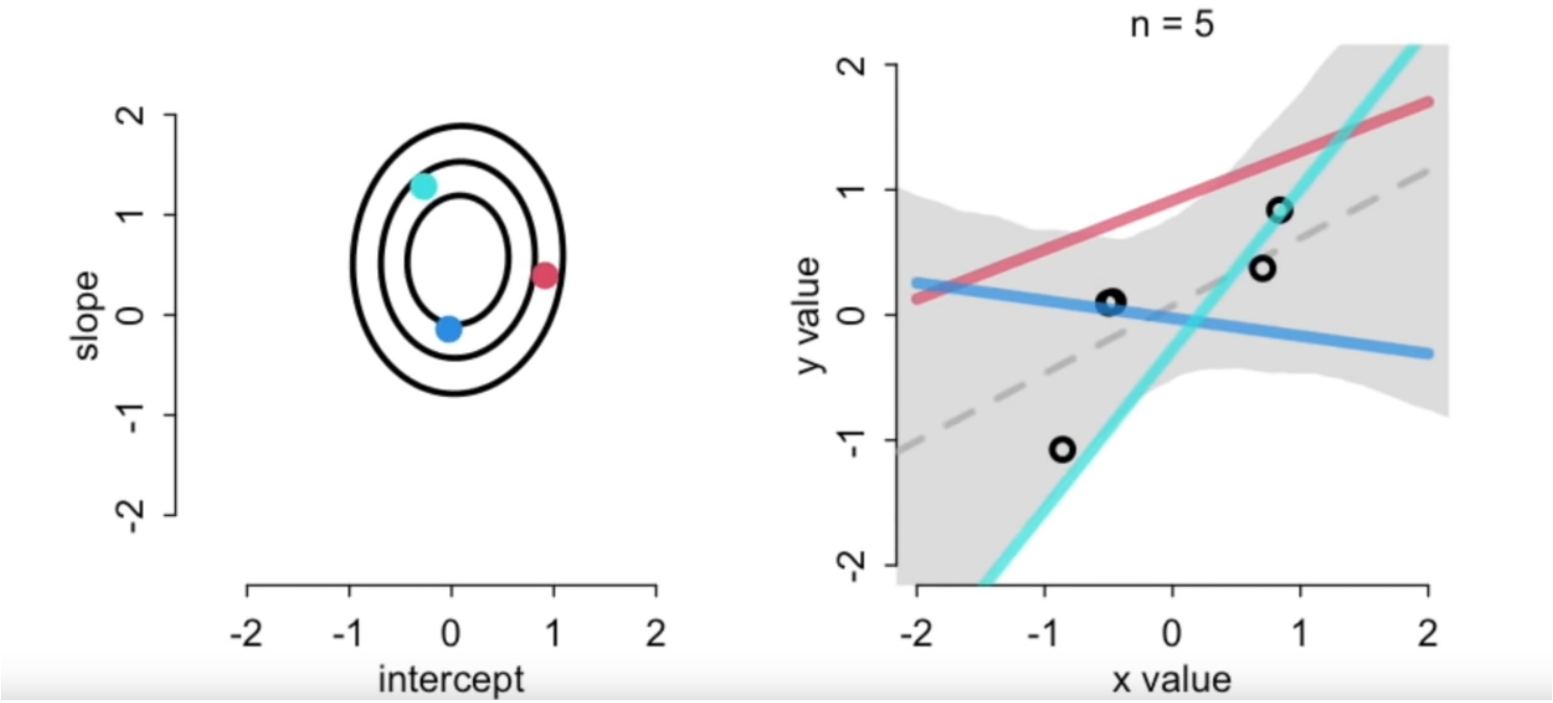
Updating the posterior



Grey regions $\approx$ highest posterior density line regions.

Statistical model

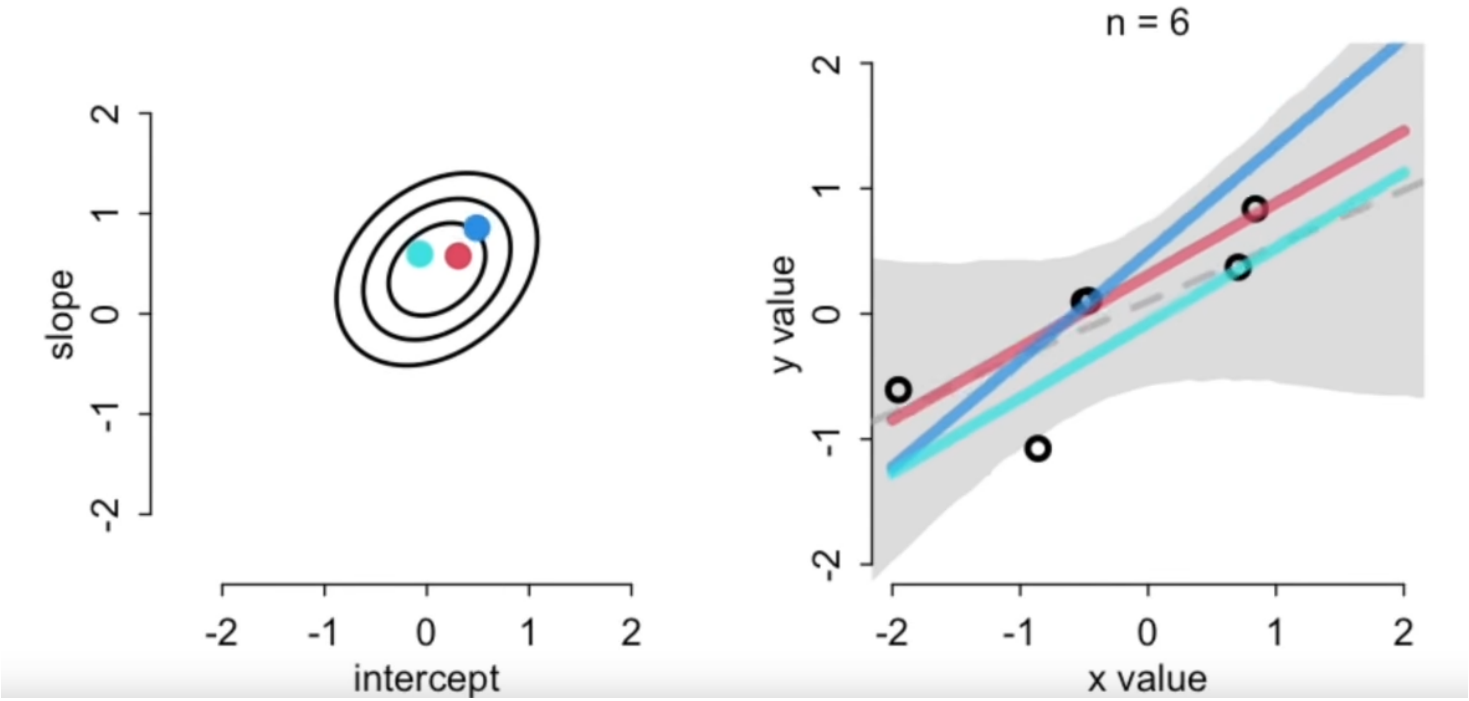
Updating the posterior



Grey regions ≈ highest posterior density line regions.

Statistical model

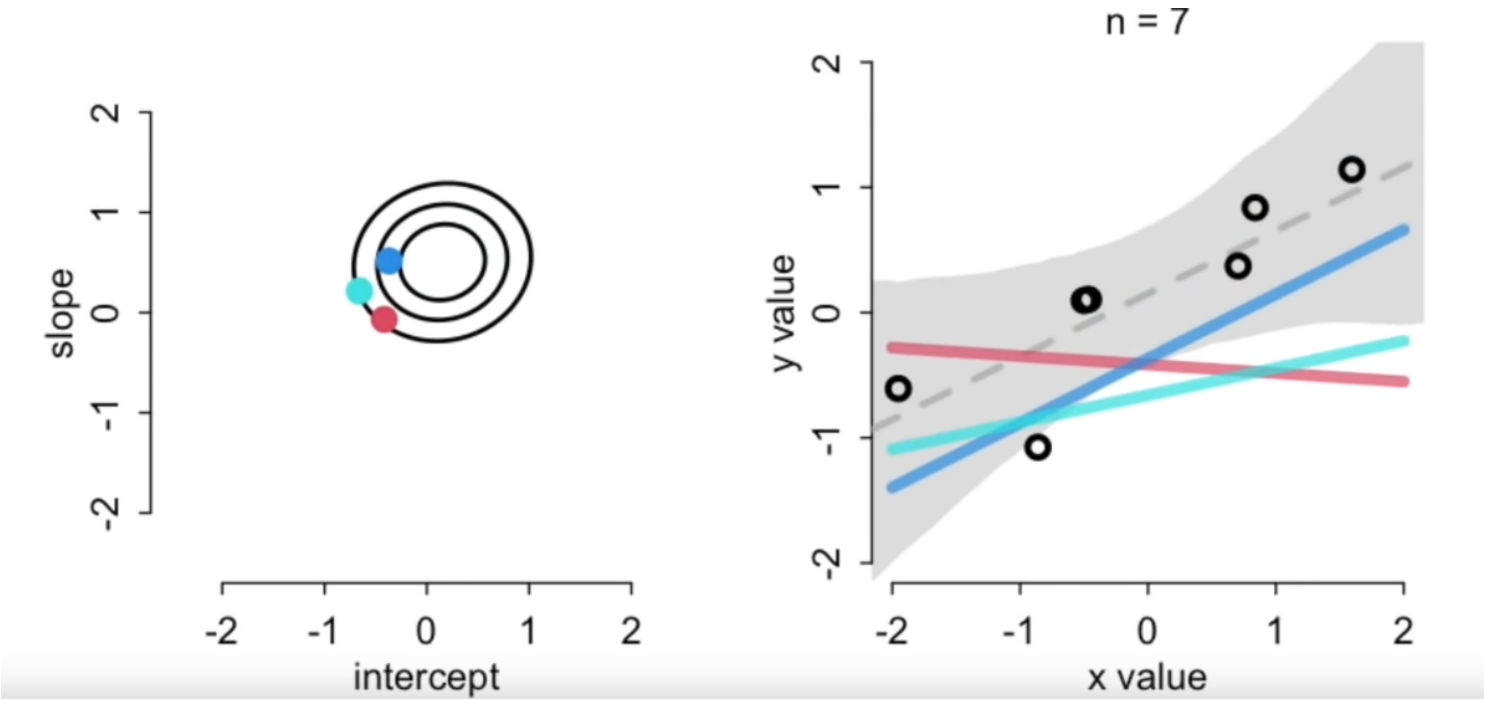
Updating the posterior



Grey regions ≈ highest posterior density line regions.

Statistical model

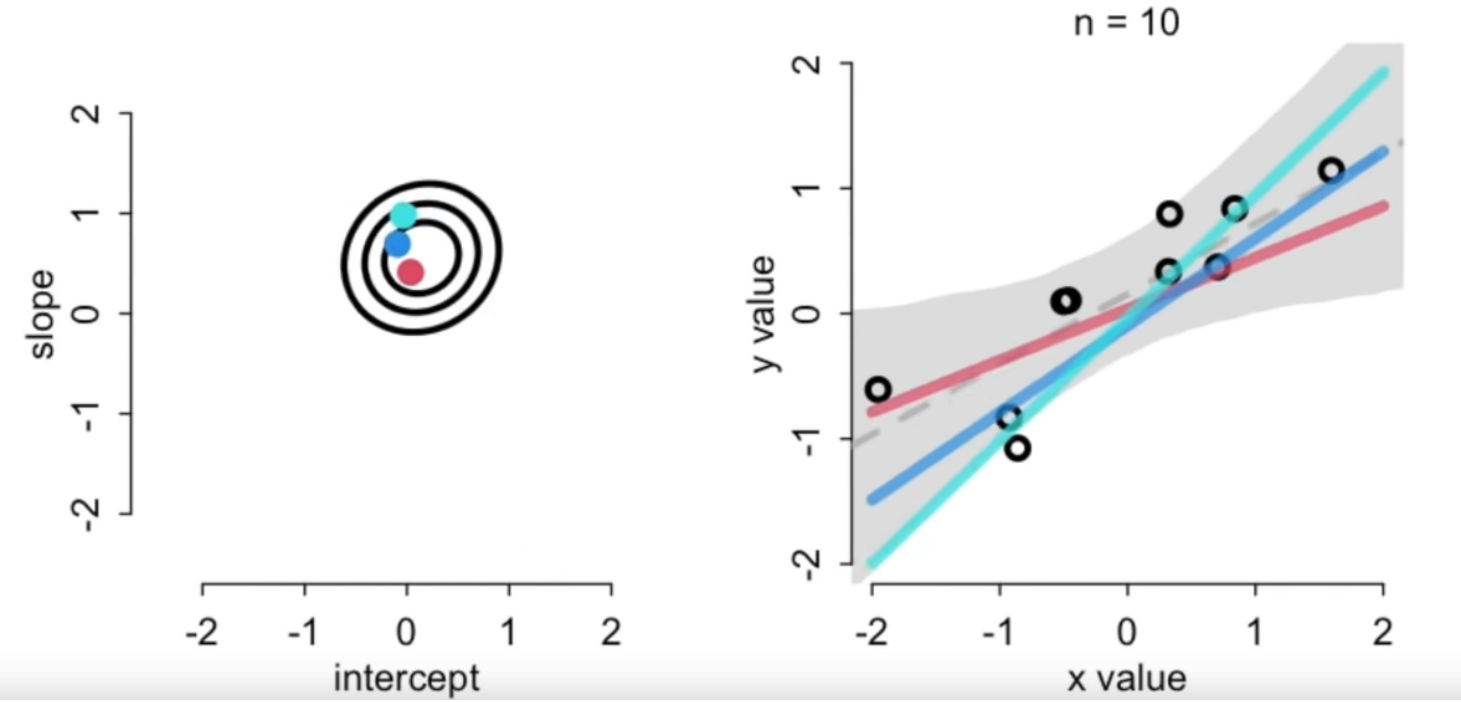
Updating the posterior



Grey regions ≈ highest posterior density line regions.

Statistical model

Updating the posterior



Grey "bowtie" regions ≈ highest posterior density line regions.

Thinking again about the $H \rightarrow W$ model

$$
\begin{aligned}
W_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta(H_i - \bar{H}) \\
\alpha &\sim \text{Normal}(?, ?) \\
\beta &\sim \text{Normal}(?, ?) \\
\sigma &\sim \text{Unif}(0, ?)
\end{aligned}
$$

The structure of this statistical model is similar to that of the generative model, but:

- Useful to re-scale the variables
- Must think about the priors

These things go together.

Rescaling

Rescaling height so that the intercept makes sense: intercept is the expected weight when the individual is of average height.

$$
\begin{aligned}
W_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta(H_i - \bar{H})
\end{aligned}
$$

Rethinking about priors

What are scientifically reasonable priors?
α average adult weight; the 60 comes e.g. from Wikipedia information on average weight in Africa. The 10 s.d. is to make the prior weakly informative and allow so generous variation.

$$
\begin{aligned}
W_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta(H_i - \bar{H}) \\
\alpha &\sim \text{Normal}(60, 10) \\
\beta &\sim \text{Normal}(?, ?) \\
\sigma &\sim \text{Unif}(0, ?)
\end{aligned}
$$

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Rethinking about scientifically reasonable priors, attempt 1

α average adult weight; 60 comes e.g. from Wikipedia information on average weight in Africa. The 10 s.d. translates prior weakly informative; allows generous variation.

β (slope) has units: kg per cm. Easy interpretation. 0: neutral direction of the relationship, and 10 shows our uncertainty about that information.

The maximum of σ is 10, which is relatively large.

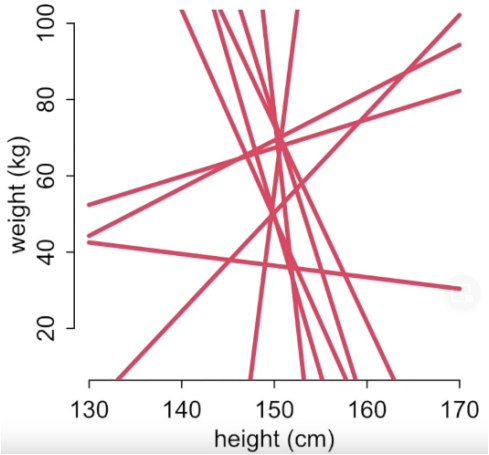
The model should learn. the parameters from the data, starting from these priors.

$$
\begin{aligned}
W_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta(H_i - \bar{H}) \\
\alpha &\sim \text{Normal}(60, 10) \\
\beta &\sim \text{Normal}(0, 10) \\
\sigma &\sim \text{Unif}(0, 10)
\end{aligned}
$$

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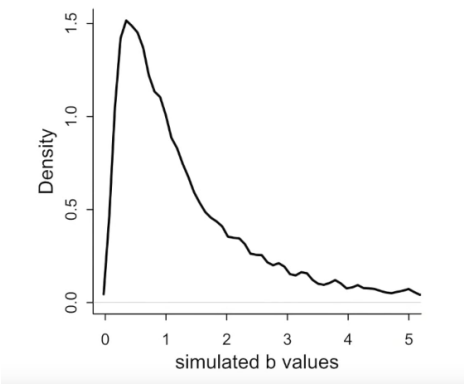
Sampling regression lines, attempt 1

```
n<-10; alpha <- rnorm(n, 60, 10); beta <- rnorm(n, 0, 10)
Hbar <- 150; Hseq <- seq(from = 130, to = 170, len = 30)
plot(NULL, xlim = c(130, 170), ylim = c(10, 100), xlab =
   "height (cm)", ylab = "weight (kg)")
for(i in 1:n){
lines(Hseq, alpha[i] + beta[i]*(Hseq-Hbar),
   lwd = 3, col = 2)}
```



Mmm, not happy!

Rethinking about scientifically reasonable priors, attempt 2



(see SR book for simulation)

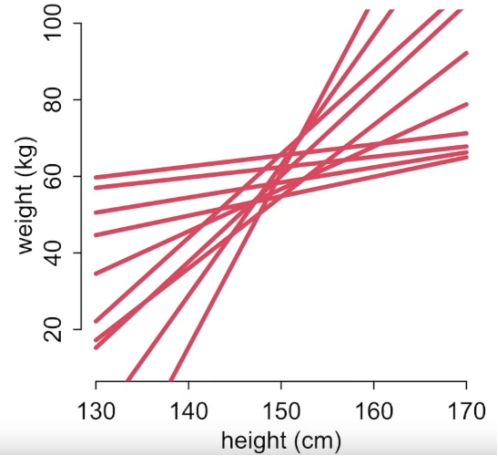
LogNormal distribution is such that if you take the log of the values are going to be normal. Or if you exponentiate normally distributed values, you get this LogNormal distribution.

$$
\begin{aligned}
W_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha + \beta(H_i - \bar{H}) \\
\alpha &\sim \text{Normal}(60, 10) \\
\beta &\sim \text{LogNormal}(0, 1) \\
\sigma &\sim \text{Unif}(0, 10)
\end{aligned}
$$

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Sampling regression lines, attempt 2

```
n<-10; alpha <- rnorm(n, 60, 10); beta <- rlnorm(n, 0, 10)
Hbar <- 150; Hseq <- seq(from = 130, to = 170, len = 30)
plot(NULL, xlim = c(130, 170), ylim = c(10, 100), xlab =
   "height (cm)", ylab = "weight (kg)")
for(i in 1:n){
lines(Hseq, alpha[i] + beta[i]*(Hseq-Hbar),
   lwd = 3, col = 2)}
```



Much better. Still some 'extreme' behaviour, but we don't want the prior to determine the relationship.

A final thought on priors

- No 'right' prior, only scientifically (or statistical) justifiable priors,
- Justify information about the prior outside the data. Like the rest of the model.
- Priors not so important in simple linear regression. Data will really dominate even the worst possible priors.
- But we need to practice now: simulate, understand, expand. Because some priors can damage the inference. Wait until multi-level models. Or neural networks. Priors will play a "similar" role to the data.

Fitting the posterior $P(\alpha, \beta, \sigma | W, H)$ model

From the likelihood and the priors (show them when we wrote the model!)

$$
\begin{aligned}
P(\alpha, \beta, \sigma | W, H) \quad & \propto \text{Normal}(W | \mu, \sigma) \\
& \times \text{Normal}(\alpha | 60, 10) \\
& \times \text{LogNormal}(\beta | 0, 1) \\
& \times \text{Uniform}(0, 10)
\end{aligned}
$$

Grid approximation is very expensive, see the SR book, 1 million computations.

Learning the parameters with the first 30 adults form Howell1

See the `lecture10-30adultsHowell-LR.mp4` video (on Stream)

Killing two birds with one stone

Validating the model and analysing the data with the same code

- Simulation-based validation
  - Bare minimum: test the model with observations simulated from the statistical model you built. Check if model/learning is broken.
  - Stronger test: simulation-based calibration. Verify that numerical algorithms are behaving as expected in the Bayesian context.
- Model formula

```
flist <- alist(
weight ~ dnorm(mu, sigma)
mu = alpha + beta*(H-Hbar)
alpha ~ dnorm(60, 10)
beta ~ dlnorm(0,1)
sigma ~ dunif(0,10)
)
```

Really looks like the maths/stats formula. You should know how to translate from from the other.

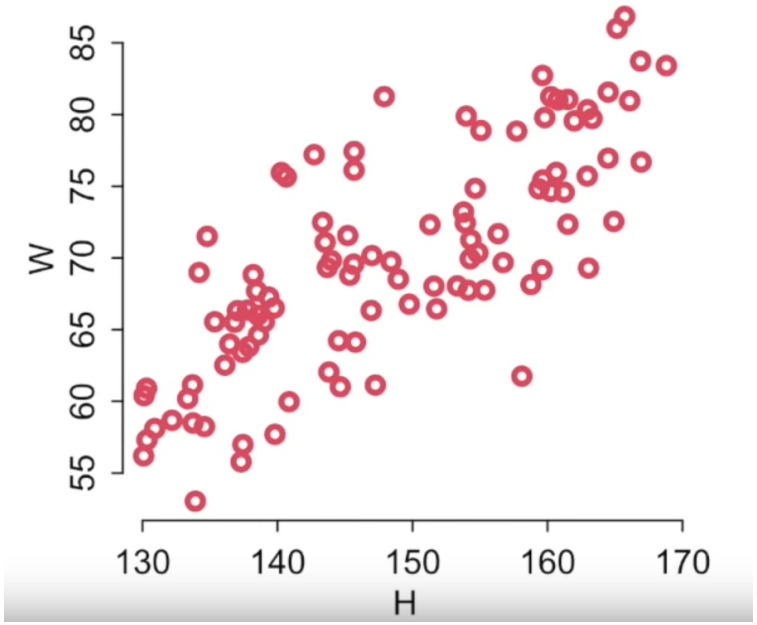
Validate with simulations

```
alpha <- 70
beta <- 0.5
sigma <- 5
n_indiv <- 100
H <- runif(n_indiv, 130, 170)
mu <- alpha+beta*(H-mean(H))
W <- rnorm(n_indiv, mu, sigma)

dat <- list(H = H, W = W,
    Hbar = mean(H))

m_validate <- quap(flist,
    data = dat)

precis(m_validate)
```



The statistical model is working. Execute it a few times and look at the output of `precis`!

Analyse the real data

The only thing that changes is how you define `dat`, from the `Howell1` data set.

```
d <- Howell1
d2 <- d[d$age>18,]
m4.3inv <- quap(alist(
    W ~ dnorm(mu, sigma),
    mu <- alpha + beta*(H-Hbar),
    alpha ~ dnorm(60, 10),
    beta ~ dlnorm(0,1),
    sigma ~ dunif(0,10)),
    data = list(W = d2$weight, H = d2$height,
       Hbar = mean(d2$height))
)
```

And you can check the outputs via table(s) or plotting simulations.

Parameter distribution and posterior simulation

Careful: the parameters are not independent: their values are intertwined and they impact the predictions together.

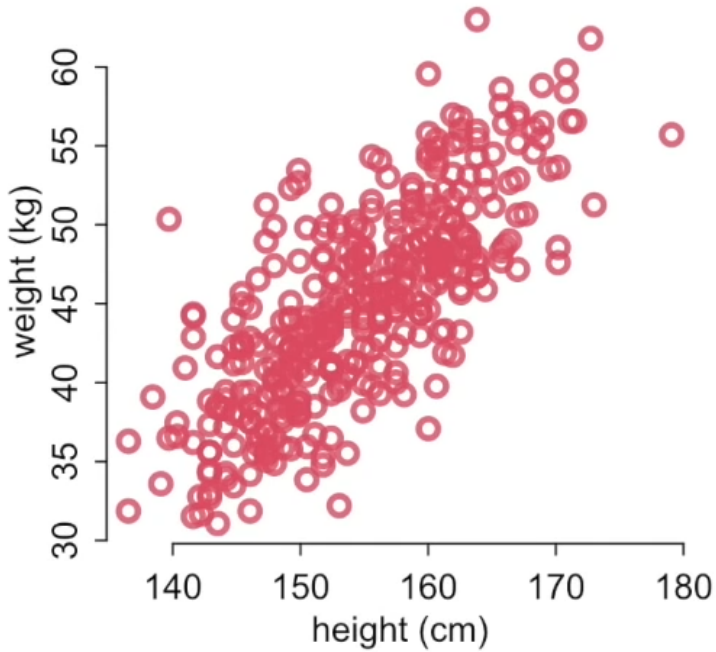
```
precis(m4.3inv)
        mean   sd  5.5% 94.5%
alpha 45.05 0.23 44.69 45.42
beta   0.62 0.03  0.58  0.67
sigma  4.24 0.16  3.99  4.50

posterior <- extract.samples(m4.3inv, n = 10000)
head(posterior)
      alpha      beta    sigma
1 45.17899 0.5622762 4.192472
2 45.19750 0.5915682 4.282521
3 45.57446 0.6343535 4.326758
4 45.32900 0.6576114 4.289662
5 45.31588 0.6230867 4.295518
6 45.32558 0.5634100 4.337623
```

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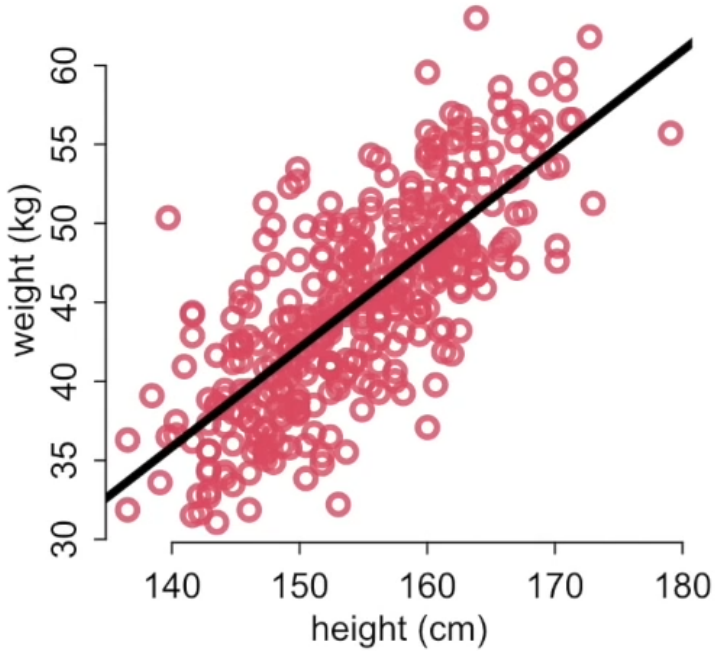
Posterior predictive distribution

1. Plot the data (the samples)



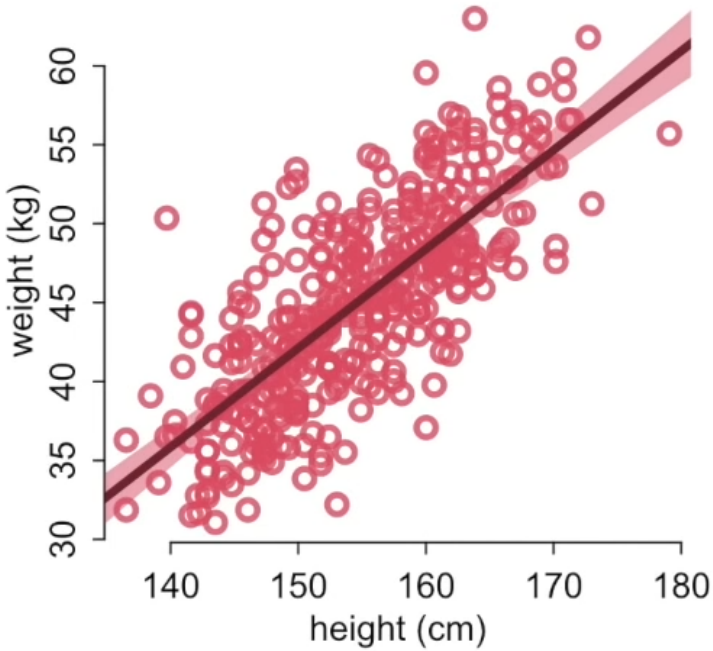
Posterior predictive distribution

1. Plot the data (the samples)
2. Plot the posterior mean



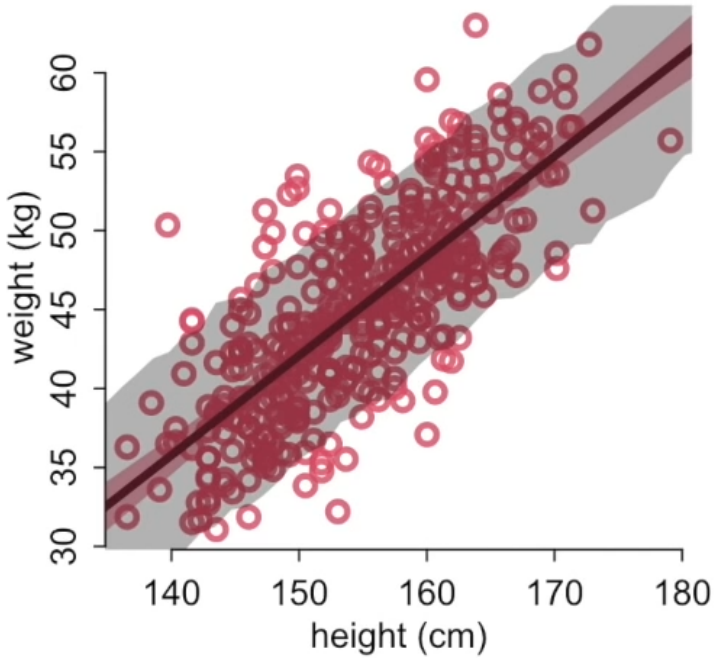
Posterior predictive distribution

1. Plot the data (the samples)
2. Plot the posterior mean
3. Plot uncertainty of the mean



Data, posterior and posterior predictive distribution

1. Plot the data (the samples)
2. Plot the posterior mean
3. Plot uncertainty of the mean
4. Plot uncertainty of the predictions



Data, posterior and posterior predictive distribution

Code to do this

Also see Section 4.4 in the R book. See the use of `link()`.

```
# plot sample
col2 <- col.alpha(2,0.8)
plot( d$height , d$weight , col=col2 , lwd=3 ,
   cex=1.2 , xlab="height (cm)" , ylab="weight (kg)" )

# expectation with 99% compatibility interval
xseq <- seq(from=130,to=190,len=50)
mu <- link(m0,data=list(H=xseq,Hbar=mean(d$height)))
lines( xseq , apply(mu,2,mean) , lwd=4 )
shade( apply(mu,2,PI,prob=0.99) , xseq ,
   col=col.alpha(2,0.5) )

# 89% prediction interval
W_sim <- sim(m0,data=list(H=xseq,Hbar=mean(d$height)))
shade( apply(W_sim,2,PI,prob=0.89) , xseq ,
   col=col.alpha(1,0.3) )
```

Recap

- Models are flexible thermometer.
- We have seen how to build a simple linear regression.
- We have seen how to fit it with a quadratic approximation.
- All we are saying is that weight is a function of height. We look at this association.
- Soon, you will be at ease reading and creating these models to answer *your* questions :)